Hassan Almosa

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PROFESSIONAL SUMMARY

Computer Science (Artificial Intelligence) graduate who builds machine learning systems for unstructured and streaming data. Delivered an LLM-based document extraction and entity resolution platform for a an industry research funder, benchmarked online anomaly detection over a Kafka streaming pipeline, and built end-to-end ML pipelines for drug discovery. Interdisciplinary background in chemistry and biosciences. Bilingual in Arabic and English. Seeking machine learning and software engineering roles with room to work close to the hard part of the problem, and where cross-disciplinary thinking add value.

PROFESSIONAL EXPERIENCE

Independent Projects & Research
Dammam, Saudi Arabia *May 2026 – Present*

- Designing and building an AI based SaaS platform for customs and logistics compliance for the logistics industry.
- Conducting exploratory research into medical AI applications in ophthalmology in Saudi Arabia's population.

Research & Development Intern, NovitaTech
Adelaide, Australia *August 2025 – January 2026*

- Sole developer of the Historical Grant Data Capture & Impact Analysis platform (detailed under Projects), an end-to-end platform that utilises AI and RAG with semantic embeddings. The project featured building a front-end intelligent human-in-the-loop raw data extraction and structuring, research tracking and insights dashboard that replaced a manual, time-intensive review process.
- Researched and planned data mining and knowledge discovery methods across organisational databases to improve operational efficiency.

IT & Managerial Assistant, Eastern Constructions
Dammam, Saudi Arabia *January 2019 – August 2021*

- Administered company IT infrastructure: Windows Server and workstation provisioning, Active Directory accounts and permissions, LAN/Wi-Fi design, and router, switch, and firewall configuration across office and project sites.
- Maintained relational databases for project, payroll, and financial records; wrote SQL queries for management reporting and implemented scheduled backup and recovery.
- Handled Microsoft 365 and SharepPoint administration, endpoint security, and all end-user support; digitised manual accounting, payroll, and tax-compliance workflows.

PROJECTS

Capstone: Hybrid Explainable AI (xAI) Quantitative Structure–Activity Relationship (QASR) System for Neuropsychiatric Drug Discovery

- Built and deployed a machine learning pipeline predicting TAAR1 receptor ligand activity, combining traditional QSAR modelling with symbolic logical inference in Prolog. Processed pharmaceutical databases of approximately 3,000 compounds through molecular featurisation pipelines, generating 3D conformers and computing structural descriptors and pharmacophore patterns.
- Implemented and compared ensemble models (Random Forest, XGBoost), a multilayer perceptron, and directed message-passing graph neural networks, integrated with a knowledge-based reasoning engine and automated experiment tracking for drug candidate prioritisation.

MIRSAD: AI Decision Layer for Air-Cargo Storage

- Built a decision layer for an air-cargo storage area (AgentX hackathon, SAL) on a SimPy simulator, with deterministic placement, a 48-hour occupancy forecast, and a LightGBM dwell model. System One (Jev) sets each action's autonomy level, behind a fail-closed rules gate, and the LLM never produces numbers. In a paired 8-seed simulation, relocations per 100 shipments fell from 303 to 140 and dig-outs from 71% to 14%.
- Delivered a FastAPI service with live simulation, a React dashboard, a three.js digital twin, a HUMAIN-backed chat agent, and an integration layer for the air-cargo operations system's API.

RAG-Based Document Intelligence Platform for Historical Research Foundation Grants Data

- Built a five-stage extraction pipeline converting 50 years of unstandardised grant records, spanning roughly 1,000 awards across formats and sources, into a unified queryable database. Utilised a locally hosted VLLM for OCR extraction.
- Implemented semantic embeddings in ChromaDB for retrieval, entity resolution utilising academic platforms' APIs queries (DOI, PubMed, ORCID, etc.) matching against researcher and publications identifiers paired with fuzzy matching, and implemented a RAG completion stage that recovered missing fields from retrieved context.
- Delivered a knowledge graph linking researchers, grants, publications and institutions, giving the organisation impact analysis via an analytics dashboard, researcher tracking and relational search over records that had no searchability before.

Real-Time IoT Anomaly Detection over Streaming Telemetry

- Built a streaming intrusion detection prototype on the Edge-IIoTset benchmark (2.2M network flow records, 14 attack types), replaying traffic through Apache Kafka into Spark/Quix, with River applying online encoding and scaling before each record reached the detector. Prometheus and Grafana captured per record latency, throughput and anomaly rates. Docker and Kubernetes to simulate streams and container management.
- Implemented, tuned and benchmarked the deep learning detectors (DeepANT, MemStream, LSTM-Autoencoder) against online tree-based methods (Half-Space Trees, Robust Random Cut Forest) and clustering baselines under identical stream conditions.

Model Context Protocol (MCP) Server

- Built and deployed a secure, serverless Model Context Protocol server on Cloudflare Workers, exposing analytical and forecasting tools over a large national economic dataset to LLM clients. Implemented OAuth authentication, a PostgreSQL data layer, and third-party API integrations.

Blockchain-based Pharmaceutical Supply Chain Solution

- Engineered a full-stack pharmaceutical supply chain tracking system on Hyperledger Fabric, using a permissioned network for smart contract–based monitoring between stakeholders. Built with Dockerised chain nodes, a hybrid Go and Node.js backend, microservices with Redis caching, role-based access control, and secure REST APIs for end-to-end traceability.

WasteID: Recycling Assistant

- Led development of a recycling guidance application using barcode and image recognition, working within an Agile framework and managing stakeholder relationships.

Engineers Without Borders Challenge

- Led a multidisciplinary team designing sustainable approaches to mitigate riverbank erosion in Queensland's *Djabugandji* community.

EDUCATION

Bachelor of Computer Science (Artificial Intelligence)
Flinders University, Adelaide, Australia **|** GPA: 6.36/7 *February 2023 – January 2026*

Bachelor of Science (Chemistry) — [Partial]
La Trobe University, Melbourne, Australia *February 2016 – November 2018*

Diploma of Bioscience
La Trobe University, Melbourne, Australia *February 2014 – November 2015*

SKILLS

- **Languages:** Python, C, C++, Go, Java, TypeScript/JavaScript, SQL, Prolog, Haskell
- **Machine Learning:** PyTorch, scikit-learn, XGBoost, River, PySAD, pandas, NumPy, RDKit, Hugging Face Transformers, MLX; neural networks, CNNs, LSTMs, autoencoders, graph neural networks, SVMs, supervised & unsupervised learning, online & incremental learning, concept drift adaptation, anomaly detection, time-series forecasting, feature engineering, transfer learning, data augmentation, cross-validation, MLOps (MLflow, DVC), expert systems & logic programming
- **Generative AI & LLMs:** prompt engineering, RAG, agentic AI workflows (tool calling, deep research agents), LangChain, LLM fine-tuning (PEFT/LoRA), RLHF, transformer architectures, text embeddings, tokenization, LLM evaluation & benchmarking, chatbots & summarisation, Hugging Face, NVIDIA NIM, Triton Inference Server, trustworthy AI (bias mitigation, data privacy & consent)
- **Data Engineering & Streaming:** Apache Kafka, Quix, Spark Structured Streaming, ETL pipeline design, windowed stream processing, real-time inference pipelines, Prometheus, Grafana
- **Data & Retrieval:** PostgreSQL, ChromaDB, vector databases, knowledge graphs, RAG architectures, semantic embeddings, entity resolution, LLM-based document extraction, local & open-source LLM deployment, exploratory data analysis
- **Web & Systems:** React, Next.js, Node.js, Express.js, Flask, FastAPI, REST APIs, Tailwind CSS, PHP, ORMs, IAM & SSO, system administration, networking, cybersecurity
- **Cloud & Tooling:** CI/CD (GitHub Actions), AWS, Azure, GCP (BigQuery, Vertex AI, Cloud Run), Cloudflare Workers, Vercel, Docker, Kubernetes (GKE), Git, Linux, n8n & Power Automate, LaTeX
- **Professional:** Leadership, project management, technical documentation, statistics, cross-functional collaboration

LANGUAGES

Arabic (Native) | **English** (Fluent)

AWARDS, SCHOLARSHIPS & COMPETITIONS

AgentX National AI Hackathon — 3rd Place Winner out of 173 Teams (2026)
Sole Developer & Architect in a Cross-Functional Team — Logistics & Ports Track, SAL Air Cargo Operations Challenge

Chancellor's Letter of Commendation (2025)
Flinders University

Chancellor's Letter of Commendation (2023)
Flinders University

"Go Beyond" Excellence Scholarship (2023)
Flinders University

Custodian of the Two Holy Mosques Excellence Scholarship Program (2014)
Ministry of Higher Education, Saudi Arabia

CERTIFICATIONS

NVIDIA Certified Associate Generative AI Large Language Models (2026)
NVIDIA, sponsored by Saudi Authority for Data and Artificial Intelligence

MLOps & AI Deployment (2026)
National Information Technology Academy

Cisco Certified CyberOps Associate (2023)
Cisco Systems